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Roll No

CE-5005(3)-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Environmental Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. The following data shows the population of a town from year 1931 to 1981. Estimate the population of town in year 2021 by any two methods and compare the results 14

Year	1931	1941	1951	1961	1971	1981
Population	52,000	65,000	90,500	1,24,000	1,64,000	2,01,000

2. Calculate the diameter and discharge of a circular sewer laid at a slope of 1 in 400 when it is running half full, and with a velocity of 1.9 m/sec. 14

Assume $n = 0.012$ in Manning's formula

3. Write note on following in detail: 14
- Fire demand
 - Forces acting on sewer pipes
4. a) Explain briefly the following processes 7
- Break point chlorination
 - Super chlorination

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- b) Give the flow diagram for 'the activated sludge process', and describe the working of the activated sludge plant. 7
- 5. a) Draw a neat sketch of a rapid sand filter and discuss the design aspect of the same. 7
- b) Differentiate between aerobic and anaerobic treatment of sewage, giving major end products. Name one treatment method of each category. 7
- 6. Explain the following terms: 14
 - a) BOD and COD
 - b) Water softening
 - c) Self cleansing velocity
 - d) Manhole
- 7. Explain the following: 14
 - a) Physical and chemical tests conducted on raw water.
 - b) Intercepting trap and anti-siphonage pipe of house sewer drainage connections.
- 8. Write a short notes on: 14
 - a) Sluice valve
 - b) Grit chamber
 - c) Sedimentation
